

A Critical-Density Literature Answer for the Scalar Hermite Interpolation Problem

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1 Status

This is a lightweight *literature-implied answer*, not an original solution. The source paper is

K. Grochenig and Y. Lyubarskii, *Gabor (Super)Frames with Hermite Functions*, arXiv:0804.4613.

In Section 4.2, immediately after equation (3.2), the authors write that, for the scalar n -th Hermite window, they do not know whether the interpolation problem

$$\sum_{k=0}^n \binom{n}{k} (-\pi\mu)^k G_n^{(n-k)}(\bar{\mu}) = \delta_{\mu,0}, \quad \mu \in \Lambda^\circ,$$

has a solution $G_n \in \mathcal{F}$ when $s(\Lambda) \geq 1/(n+1)$. They note that it is conceivable that the sufficient condition $s(\Lambda) < 1/(n+1)$ is also necessary.

The supporting paper is M. Faulhuber, I. Shafkulovska, and I. Zlotnikov, *On the Frame Property of Hermite Functions and Exploration of their Frame Sets*, J. Fourier Anal. Appl. 31:21 (2025).

2 Identification

The source paper recalls the characterization from its earlier Proposition 3.3: $\mathcal{G}(H_n, \Lambda)$ is a scalar Gabor frame for $L^2(\mathbb{R})$ if and only if there is a Bessel dual window γ_n whose Bargmann transform $G_n \in \mathcal{F}$ satisfies the displayed interpolation problem above. Thus any later theorem showing that $\mathcal{G}(H_n, \Lambda)$ is a frame supplies a solution of equation (3.2).

Faulhuber–Shafkulovska–Zlotnikov prove, in their Theorem 1.1, that for

$$\Lambda_n = (1/\sqrt{n+1})\mathbb{Z}^2$$

the system $\mathcal{G}(h_n, \Lambda_n)$ is a Gabor frame for $L^2(\mathbb{R})$ if and only if $n \geq 4$. Their normalization uses $\text{vol}(\Lambda) = ab$ for $\Lambda = a\mathbb{Z} \times b\mathbb{Z}$, while the source paper's $s(\Lambda)$ is the same covolume. Therefore

$$s(\Lambda_n) = \text{vol}(\Lambda_n) = \frac{1}{n+1}.$$

Combining these two facts gives the promised critical examples: for every $n \geq 4$, equation (3.2) has a solution in the Fock space at the critical density $s(\Lambda) = 1/(n+1)$. Consequently, the proposed necessity of the strict inequality $s(\Lambda) < 1/(n+1)$ is false for the scalar Hermite interpolation problem.

3 Scope

This does not classify all lattices with $s(\Lambda) \geq 1/(n+1)$. It gives explicit square-lattice critical examples. It also does not answer the broader general M^1 superframe strict-density conjecture stated earlier in the source paper; that conjecture concerns vector-valued superframes for arbitrary M^1 windows, whereas the implication recorded here concerns the scalar Hermite window and the source’s interpolation problem (3.2).

The supporting paper is aware that it enlarges the Grochenig–Lyubarskii “safety region” and cites the Hermite-frame literature, but it does not state the source’s equation (3.2). The relation to the interpolation problem is the agent-identified consequence of the source paper’s frame/interpolation equivalence.

4 Search Notes

The run indexes were searched for arXiv:0804.4613, Hermite, superframe, polyanalytic, and critical-density keywords; no prior packet for this exact source question was found. Web searches for Hermite Gabor frame density and the Grochenig–Lyubarskii conjecture found the 2025 JFAA article, whose Theorem 1.1 is the decisive supporting theorem.

5 References

- K. Grochenig and Y. Lyubarskii, *Gabor (Super)Frames with Hermite Functions*, Math. Ann. 345 (2009), 267–286; arXiv:0804.4613.
- M. Faulhuber, I. Shafkulovska, and I. Zlotnikov, *On the Frame Property of Hermite Functions and Exploration of their Frame Sets*, J. Fourier Anal. Appl. 31:21 (2025), DOI: [10.1007/s00041-025-10153-y](https://doi.org/10.1007/s00041-025-10153-y).